

Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	2 Marks

TECHNOLOGY STACK

DEFINE YOUR TECHNOLOGY STACK:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution.

TECHNOLOGY STACK DETAILS

S.No	Architecture Component	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit Framework (Python)	Enables the rapid construction of a clean, interactive data dashboard entirely in Python, providing responsive rendering for user interaction metrics.
2	Backend / Server-Side	Python (Modular Pipeline)	Serves as the native language ecosystem for machine learning and audio processing pipelines, managing core operations seamlessly.
3	Database / Data Storage	SQLite / SQLAlchemy ORM	SQLite offers zero-configuration standalone file database storage for local analytics, with a clear upgrade pathway via SQLAlchemy.
4	Cloud / Deployment	Local Desktop Environment	Configured for isolated execution inside virtual environments, decoupled dynamically using standard environment files.
5	Version Control & CI/CD	Git / GitHub / Pytest Suite	Maintains clean code repositories while utilizing automated test runs to safeguard evaluation calculations across updates.
6	Third-Party APIs / Tools	Google Gemini, OpenAI Whisper, Sentence-BERT, Librosa	Powers the complete intelligence engine: Whisper for STT, Sentence-BERT for understanding metrics, Librosa for fluency, and Gemini API for speech critiques.